

Forecasting wheat yield eight weeks before harvest

119 districts across Haryana, Punjab and Uttar Pradesh.

A point forecast, a full probability distribution, and an early warning of collapse.

Every figure in this deck is machine-verified — 96 checks, 96 pass

THE TASK

Predict the harvest before it happens

5 March

the forecast date — eight weeks before harvest

119

districts, averaging 3,400 km² each

4,500

kg per hectare — the typical yield

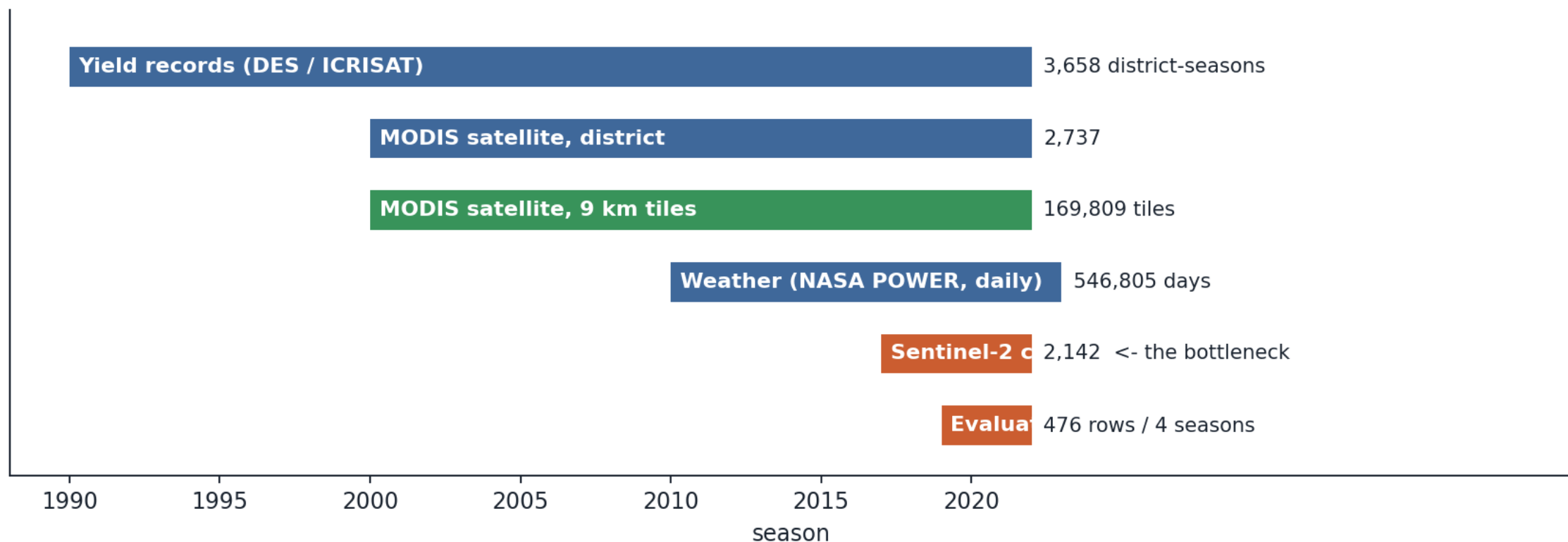
Why it is hard

- The crop is not finished — a late-March heatwave can still destroy 20% of it
- Districts are huge — half can fail while the other half thrives
- Ground truth arrives once a year, and neighbouring districts fail together

The sharpest data starts too late

What data exists, and when it starts

Sentinel (2017) forced a 4-season evaluation; MODIS (2000) allows 19



Sentinel-2 begins in 2017, leaving four testable harvests. MODIS reaches back to 2000 — nineteen.

Five stages, each a small correction

typical error, kg/ha →

0	Baseline	Weighted average of the last three harvests	333
1	The Blend	Five simple models, a disagreement gate, movement calibration	273
2	Weather ahead	What changes when the model may see the 10-day forecast	272
3	Training data	Train on more seasons (small, uncertain)	—
4	Crop vision	A transformer reads satellite crop condition	265

Never let a new component predict the harvest alone. Let it nudge a conservative estimate.

A network that never sees a harvest figure

The problem Thousands of examples of how wheat develops. Only a handful of harvests.

The solution Train it on the data-rich question — how will the crop change between January and March — then hand its internal summary to a simple model for the data-poor one.

Attention, in one line

Each piece of input decides how much to listen to every other piece — then takes a weighted average.

67,359

learned numbers — one layer of a small language model

16

values it hands to the yield model, down from 30

± 130

kg/ha — how far it can move a forecast

Removing inputs made it stronger

Before

30

- 6 raw satellite averages needing no network at all
- 8 weak change summaries
- 16 genuine network outputs

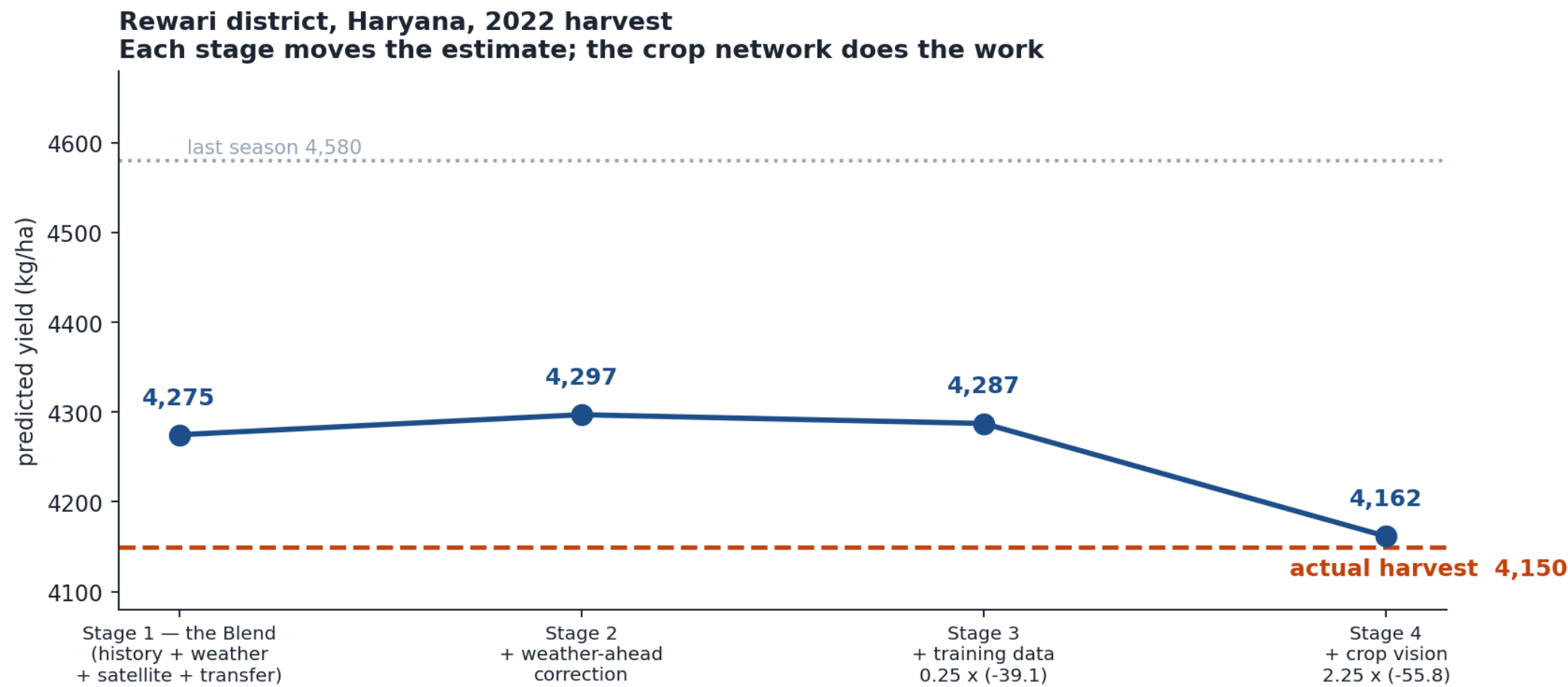
After

16

- The 6 raw averages correlated 0.024 with the model's errors — nothing
- They made accuracy worse
- The remaining 16 could then be trusted at nearly double the weight

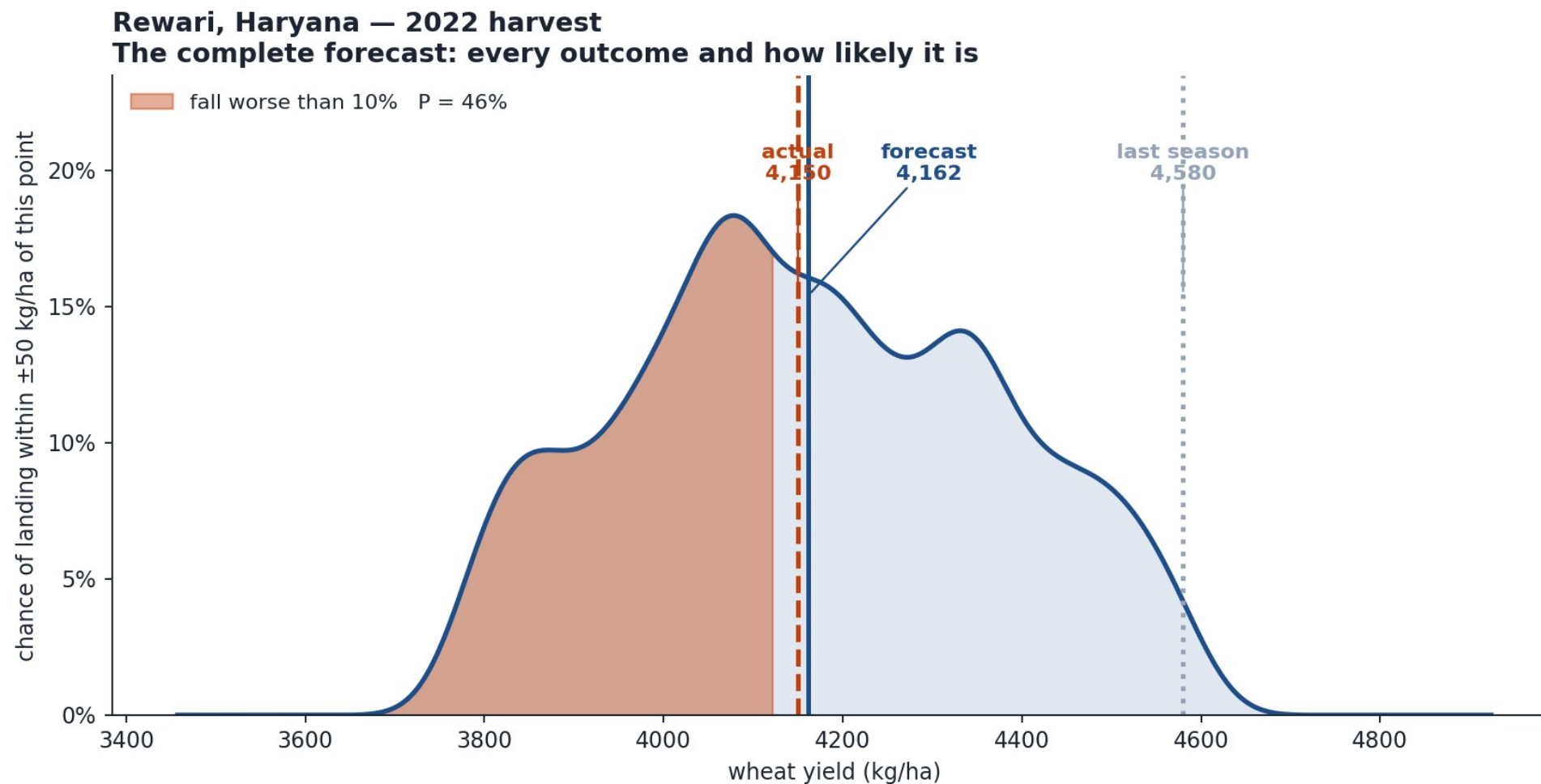
Removing features let the remaining signal be believed harder.

Rewari district, Haryana — 2022

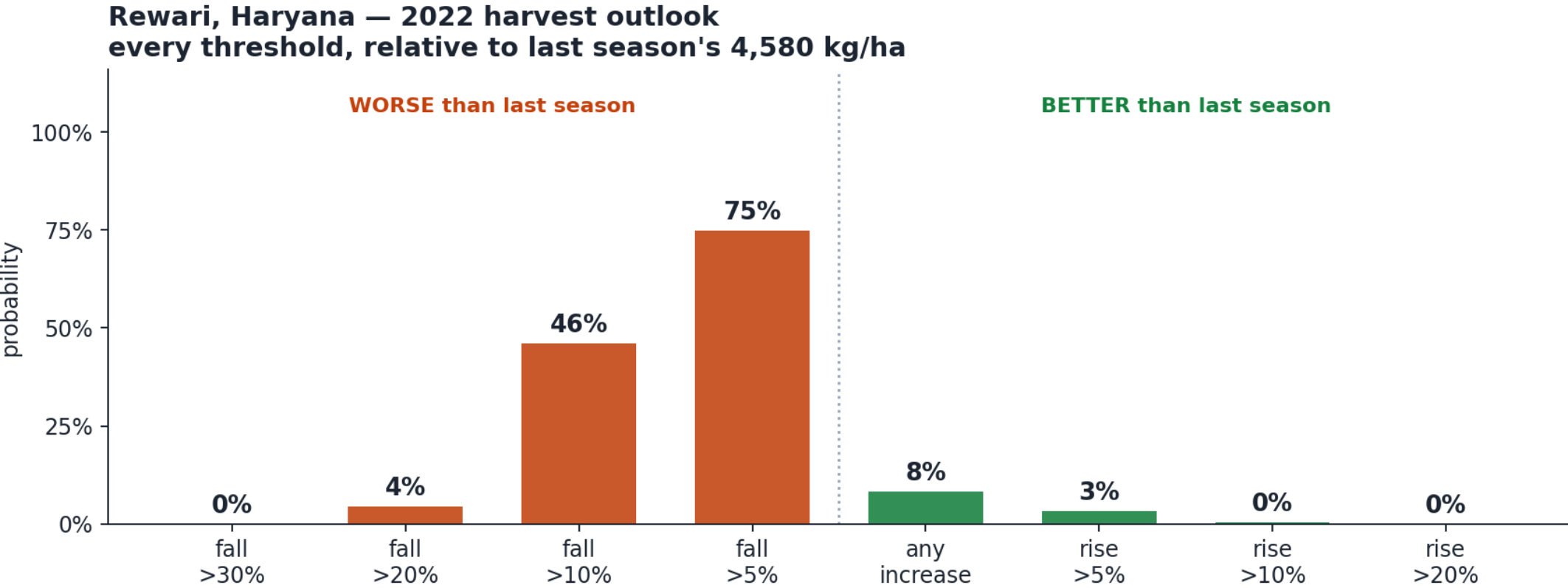


WORKED EXAMPLE

The forecast is a distribution, not a number



Every threshold a planner asks about



Rewari fell 9.4%. The model called a decline 92% likely and put it right at the 10% boundary.

DOES IT WORK?

The probabilities are honest

When the model says...	Stated	Actually happened
Fall worse than 5%	29.9%	29.2%
Fall worse than 10%	11.5%	10.9%
Fall worse than 20%	1.2%	2.3%
Any increase	45.9%	43.5%

Across all 476 district-seasons

80%

ranges cover

80.9%

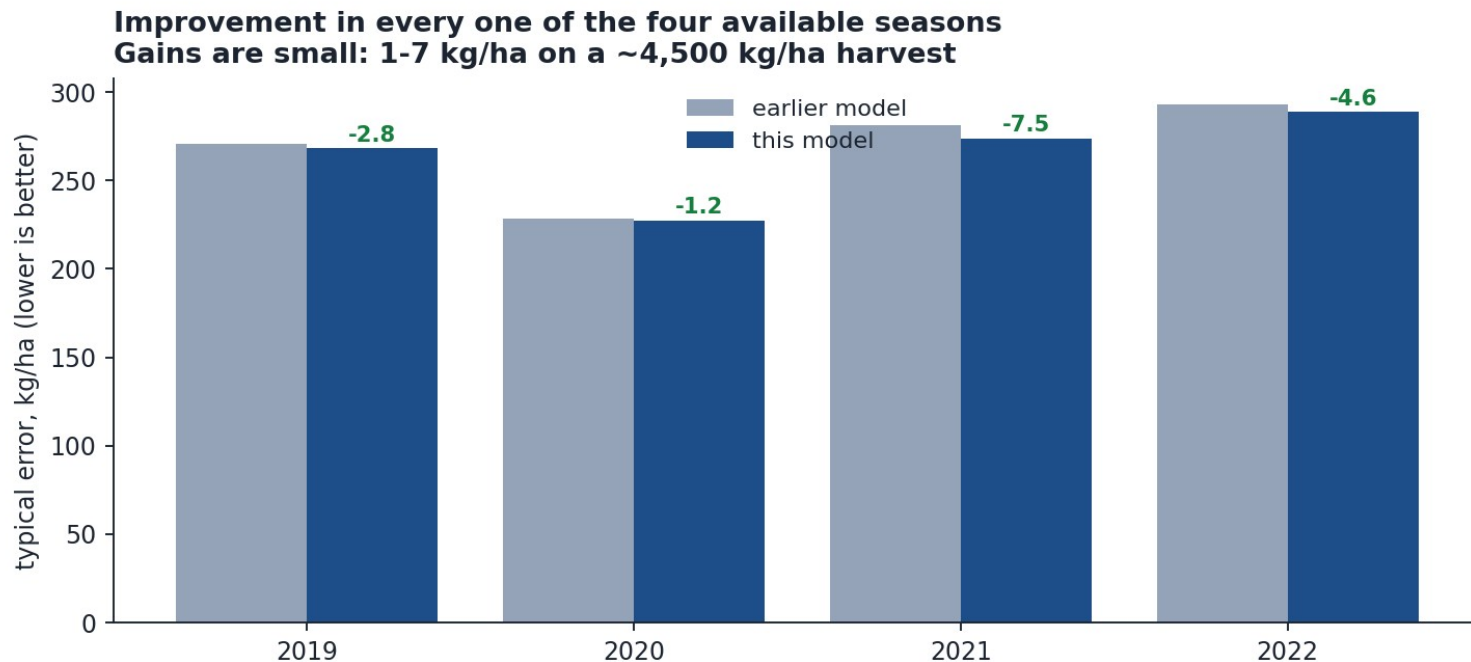
of the time

A 10% chance
happens 10% of the time.

Slightly over-confident about catastrophes: 1.2% stated for falls worse than 20%, where 2.3% occur.

DOES IT WORK?

Better in every season available



265

kg/ha typical error — 5.9% of a harvest

4 of 4

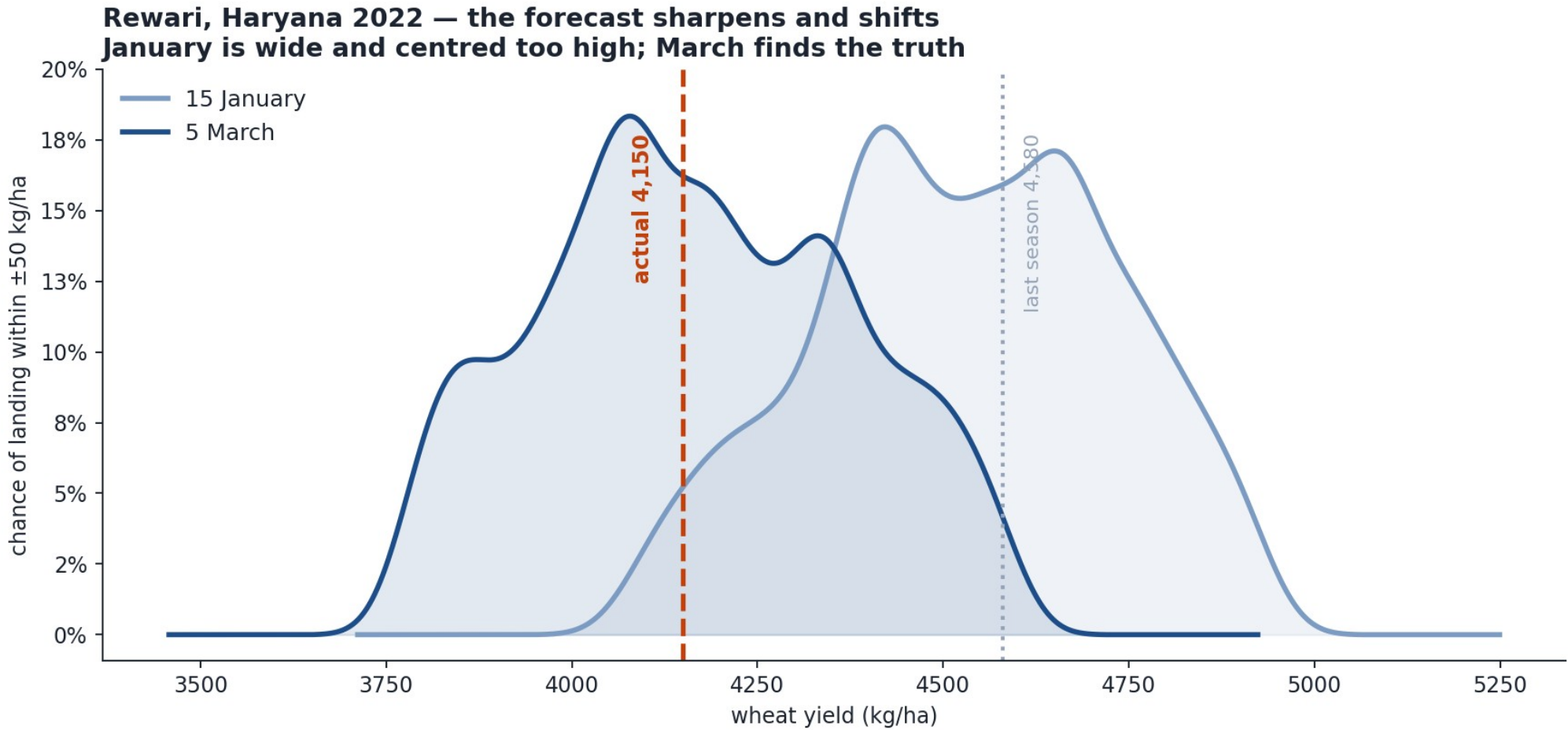
seasons improved

78.8%

up-or-down called correctly

Never “100% confident” — with four seasons a resampling test reports certainty automatically whenever all four agree. Say “4 of 4” and let the reader judge.

What waiting until March buys you



The same district, seven weeks apart

	15 January	5 March
Point forecast	4,555	4,162
Error	405	12
P(increase)	43%	8%
P(fall over 10%)	9%	46%

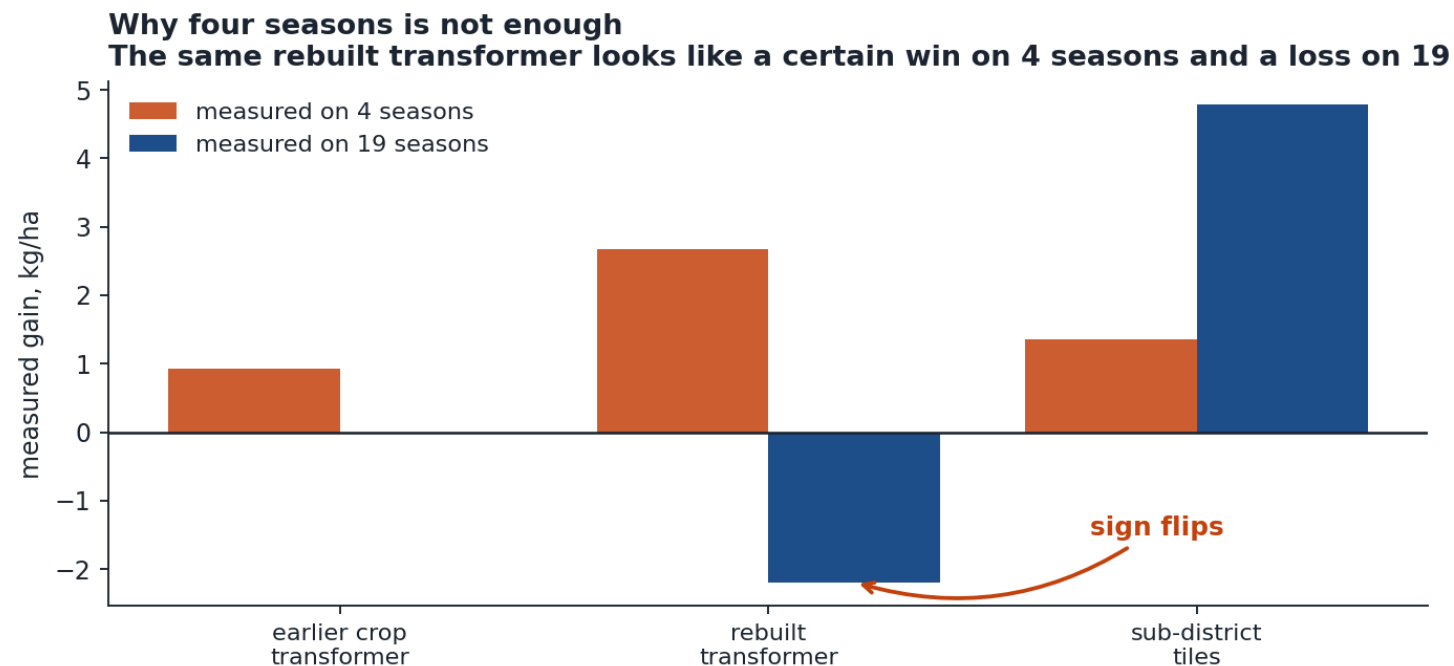
4,150

the actual harvest

January said a serious decline was 9% likely.
March said 46%.

The information arriving in February and early March is what turns a wrong answer into a right one.

Four seasons can point the wrong way



A rebuilt network looked like a certain win on four seasons and a loss on nineteen. The sign flips.

The noise floor

Re-run the model changing only the order the input columns are listed in. No information changes — but the score moves anyway.

± 0.6

kg/ha of pure noise

The earlier model's headline result was 0.9 — inside its own noise floor.

Most things did not work

Rebuilt network, better architecture

worse over 19 seasons

Sub-district tiles into the existing model

no effect

Network output as a direct correction

correlation 0.024 with real errors

Network output in the uncertainty model

coverage collapsed 79% → 60%

Fine-tuning the network directly on yield

378 kg/ha versus 265

Collapsing all five stages into one model

-77 kg/ha

Every intervention that supplied more or better data helped. Every cleverer network on the same data did not.

What it delivers

265

kg/ha typical error — 5.9% of a harvest

80.9%

of harvests land inside the 80% range

0.82

skill at ranking a >10% collapse

3 hrs

to run the whole pipeline on one laptop

The largest verified improvement came from removing fourteen useless inputs — not from adding anything.